

Reversible instance normalization

Executive summary of obtained results

Decision. Across the corrected 15-setting core grid, instance normalization is strongly beneficial relative to global standard normalization. The affine RevIN layer changes results by less than 0.1% at the median, while normalized-loss training helps mainly on Test1 and is mixed on Test2. Tables 2 and 3 contain the complete quantitative support for these conclusions.

Evidence completeness

Archived core evidence	Datasets	Settings	Methods	Seeds	Complete seed runs
Traffic, Electricity, Solar; PatchTST	3	5	6	3	270 / 270

Table 1: The 90 method configurations have non-empty metrics, dataset configuration, completion signature, and criterion plot for all three seeds. The 94 schema-1 manifests consist of these 90 configurations plus four isolated test configurations.

Aggregate normalization result

Method	Test1 gain	Test2 gain
Instance normalization, MSE	14.44%	16.50%
Instance normalization, nMSE	17.51%	16.65%
Affine RevIN, MSE	14.57%	16.58%
Affine RevIN, nMSE	17.57%	16.70%
Validation-selected method	16.20%	16.75%
Optimistic test oracle	17.72%	17.05%

Table 2: Mean relative MSE improvement over global standard normalization, equal-weighting all 15 dataset/setting rows. An instance-normalized method is best on 15/15 Test1 rows and 14/15 Test2 rows; global standard wins only Solar 336:720 on Test2.

Affine and training-loss ablations

Paired comparison over 15 settings	Test1 median	Test1 max	Test2 median	Test2 max
affine–non-affine , MSE training	0.097%	0.739%	0.069%	0.455%
affine–non-affine , nMSE training	0.073%	0.239%	0.051%	0.204%
nMSE minus MSE training, non-affine	Test1		Test2	
Mean relative effect	+4.147%		−0.019%	
Median relative effect	+0.545%		+0.314%	
Rows improved	12/15		10/15	
Worst row	−0.855%		−7.833%	

Table 3: The affine layer is numerically negligible. Normalized-loss training has a positive Test1 mean driven by Electricity but no aggregate Test2 gain; its worst Test2 row is Electricity 168:24.

Run-level uncertainty

Example	Seed 1	Seed 2	Seed 3	Mean	Sample SD
Electricity 504:168, Test1 MSE, non-affine instance	11.07M	0.261M	0.221M	3.85M	6.25M

Table 4: The run seed fixes user partition, sampling, initialization, and optimization together. Reported seed variation is therefore total run uncertainty rather than optimizer-only uncertainty.

Evidence retained before deleting artifacts

Retained conclusion	Quantitative basis
Use instance normalization	14.44–17.57% Test1 and 16.50–16.70% Test2 gain versus standard normalization in Table 2.
Do not require affine RevIN	Median paired effects are 0.051–0.097%, with all maxima below 0.74%, in Table 3.
Keep nMSE selectable	+4.147% mean on Test1 but −0.019% on Test2, with a −7.833% worst row, in Table 3.
Report total run uncertainty	The representative three-seed spread is 0.221M–11.07M and SD 6.25M in the uncertainty table.

Historical boundary. These are results of the corrected current protocol, not a resubmission or numerical reproduction of the rejected ECML/AALTD paper. The original submission remains read-only guidance in the workspace-level `latex/submissions` directory.